ENGINEERING DOCUMENTATION PORTFOLIO · VOLUME 16

Deployment, Configuration, and Operations

Configuration, containers, CI/CD, deployment, backup, and recovery.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

> **Source of truth.** Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

Operational model

Development and production containerize FastAPI with MongoDB, Redis, MinIO, and Weaviate. Production uses immutable releases, Caddy TLS/edge controls, readiness gates, smoke checks, and rollback. Backup scripts capture state with checksums.

- `/health` is liveness; `/ready` probes dependencies.
- Restore Redis rather than disabling fail-closed rate limiting.
- Copy backups off-host and perform restore drills.
- Migration failure prevents serving traffic.
- Alert rules need an on-call routing destination.

File	Responsibility
`.env.example`	# App # Generate with: python -c "import secrets; print(secrets.token_urlsafe(48))" SECRET_KEY=replace-with-a-unique-random-secret ALGORITHM=HS256 ACCESS_TOKEN_EXPIRE_MINUTES=600 ENVIRONMENT=development API_DOCS_ENABLED=true METRICS_ENABLED=true CORS_ALLOWED_ORIGINS=http://localhost:5173,http://localhost:3000 TRUSTED_HOSTS=localhost,127.0.0.1,testserver # MongoDB MONGO_ROOT_USERNAME=replace-with-local-mongo-user MONGO_ROOT_PASSWORD=replace-with-local-mongo-password MONGO_URI=mongodb://replace-with-local-mongo-user:replace-with-local-mongo-password@mongo:27017/?authSource=admin MONGO_DB=eurskem_ai # Weaviate WEAVIATE_HOST=weaviate WEAVIATE_PORT=8080 WEAVIATE_GRPC_PORT=50051 WEAVIATE_API_KEY_USER=eurskem-app WEAVIATE_API_KEY=replace-with-local-weaviate-api-key # MinIO MINIO_ENDPOINT=minio:9000 MINIO_ACCESS_KEY=replace-with-local-minio-user MINIO_SECRET_KEY=replace-with-local-minio-password MINIO_BUCKET=eurskem-ai-docs # Redis REDIS_PASSWORD=replace-with-local-redis-password REDIS_URL=redis://:replace-with-local-redis-password@redis:6379/0 # Grafana (local monitoring profile) GRAFANA_ADMIN_PASSWORD=replace-with-local-grafana-password # Runtime readiness.
`app/config.py`	Config module.
`Dockerfile`	FROM alpine:3.22 AS scientific-skills ARG SCIENTIFIC_SKILLS_REF=v2.59.0 RUN apk add --no-cache ca-certificates git \ && git clone \ --branch "${SCIENTIFIC_SKILLS_REF}" \ --depth 1 \ https://github.com/K-Dense-AI/scientific-agent-skills.git \ /scientific-agent-skills \ && rm -rf /scientific-agent-ski
`docker-compose.yml`	services: app: build: context: .
`docker-compose.production.yml`	name: eurskem-ai x-logging: &default-logging driver: json-file options: max-size: "10m" max-file: "5" services: caddy: image: caddy:2.10.0-alpine restart: unless-stopped ports: - "80:80" - "443:443" - "443:443/udp" environment: DOMAIN: ${DOMAIN:?Set DOMAIN in .env.production} ACME_EMAIL: ${ACME_EMAIL:?Set ACME_EMAIL in .env.production} MAX_REQUEST_BODY_SIZE: ${MAX_REQUEST_BODY_SIZE:-220MB} volumes: - ./deploy/ionos/Caddyfile:/etc/caddy/Caddyfile:ro - caddy-data:/data - caddy-config:/config depends_on: app: condition: service_healthy frontend: condition: service_healthy networks: - edge logging: *default-logging frontend: build: context: ./ui args: VITE_API_URL: "" restart: unless-stopped networks: - edge healthcheck: test: ["CMD-SHELL", "wget -q --spider http://127.0.0.1:8080/healthz"] interval: 20s timeout: 5s retries: 5 start_period: 10s read_only: true tmpfs: - /var/cache/nginx - /var/run - /tmp logging: *default-logging app: build: context: .

File	Responsibility
`docs/OPERATIONS_RUNBOOK.md`	Relevant implementation file.
`.github/workflows/ci.yml`	name: CI on: push: branches: ["main"] pull_request: permissions: contents: read concurrency: group: ci-${{ github.ref }} cancel-in-progress: true jobs: backend: runs-on: ubuntu-latest timeout-minutes: 30 env: ENVIRONMENT: test SECRET_KEY: ci-only-secret-key-never-use-in-production DEV_BYPASS_ENABLED: "true" MONGO_ROOT_USERNAME: eurskem-ci MONGO_ROOT_PASSWORD: ci-mongo-only-not-a-production-secret MONGO_URI: mongodb://eurskem-ci:ci-mongo-only-not-a-production-secret@localhost:27017/?authSource=admin MONGO_DB: eurskem_ai WEAVIATE_HOST: localhost WEAVIATE_PORT: "8080" WEAVIATE_GRPC_PORT: "50051" WEAVIATE_API_KEY_USER: eurskem-ci WEAVIATE_API_KEY: ci-weaviate-only-not-a-production-secret MINIO_ENDPOINT: localhost:9000 MINIO_ACCESS_KEY: eurskem-ci MINIO_SECRET_KEY: ci-minio-only-not-a-production-secret MINIO_BUCKET: eurskem-ai-docs REDIS_PASSWORD: ci-redis-only-not-a-production-secret REDIS_URL: redis://:ci-redis-only-not-a-production-secret@localhost:6379/0 GRAFANA_ADMIN_PASSWORD: ci-grafana-only-not-a-production-secret PAPER_SEARCH_MCP_ENABLED: "false" READINESS_REQUIRED_SERVICES: mongo,weaviate,minio,redis,checkpointer,mcp:eurskem steps: - uses: actions/checkout@9c091bb21b7c1c1d1991bb908d89e4e9dddfe3e0 # v7.0.0 with: persist-credentials: false - name: Install uv uses: astral-sh/setup-uv@v5 with: enable-cache: true - name: Set up Python run: uv python install 3.12 - name: Start required services run: docker compose up -d mongo weaviate minio redis - name: Wait for service health run: \| for attempt in {1..60}; do unhealthy="$(docker compose ps --format json | jq -rs ' [.[] | if type == "array" then .[] else .
`.github/workflows/deploy-ionos.yml`	name: Deploy IONOS Production on: workflow_run: workflows: ["CI"] types: [completed] permissions: contents: read concurrency: group: ionos-production cancel-in-progress: false jobs: deploy: if: >- github.event.workflow_run.conclusion == 'success' && github.event.workflow_run.event == 'push' && githu
`.github/workflows/live-llm-tests.yml`	name: Live LLM Tests on: workflow_dispatch: inputs: provider: description: Provider test to run (uses paid API calls) required: true default: openai type: choice options: - openai - anthropic - both permissions: contents: read concurrency: group: live-llm-tests cancel-in-progress: false jobs: openai
`deploy/ionos/backup.sh`	#!/usr/bin/env bash set -euo pipefail root_dir=/opt/eurskem release_dir=${root_dir}/current env_file=${root_dir}/shared/.env.production backup_root=${root_dir}/backups stamp=$(date -u +%Y%m%dT%H%M%SZ) work_dir=$(mktemp -d "${backup_root}/.backup-${stamp}.XXXXXX") archive=${backup_root}/eurskem-${stamp}.tar.gz stopped=0 cleanup() { if [[${stopped} -eq 1 && -d ${release_dir}]]; then cd "${release_dir}" docker compose --env-file "${env_file}" \ -f docker-compose.production.yml up -d fi if [[-d ${work_dir}]]; then rm -rf -- "${work_dir}" fi } trap cleanup EXIT if [[!
`deploy/ionos/cleanup_retired_workflow_artifacts.sh`	#!/usr/bin/env bash set -euo pipefail workflows_dir=${1:?workflow directory is required} if [[!
`deploy/ionos/deploy_release.sh`	#!/usr/bin/env bash set -euo pipefail release_dir=${1:?release directory is required} release_sha=${2:?release SHA is required} root_dir=/opt/eurskem shared_dir=${root_dir}/shared env_file=${shared_dir}/.env.production previous_dir="" if [[!
`deploy/ionos/init-minio.sh`	#!/bin/sh set -eu mc alias set local http://minio:9000 "$MINIO_ROOT_USER" "$MINIO_ROOT_PASSWORD" mc mb --ignore-existing "local/$MINIO_BUCKET" mc admin user add local "$MINIO_ACCESS_KEY" "$MINIO_SECRET_KEY" 2>/dev/null \|\| true sed "s/__BUCKET__/$MINIO_BUCKET/g" \ /config/minio-app-policy.json > /tmp
`deploy/ionos/minio-app-policy.json`	{ "Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Action": ["s3:GetBucketLocation", "s3:ListBucket"], "Resource": ["arn:aws:s3:::__BUCKET__"] }, { "Effect": "Allow", "Action": ["s3:GetObject", "s3:PutObject", "s3:DeleteObject"], "Resource": ["arn:aws:s3:::__BUCKET__/*"] }] }
`deploy/ionos/mongo-init.js`	const appDbName = process.env.MONGO_DB || "eurskem_ai"; const appDb = db.getSiblingDB(appDbName); if (!appDb.getUser(process.env.MONGO_APP_USERNAME)) { appDb.createUser({ user: process.env.MONGO_APP_USERNAME, pwd:

File	Responsibility
	process.env.MONGO_APP_PASSWORD, roles: [{ role: "readWrite", db: appDbName }], }); }
`deploy/ionos/restore.sh`	#!/usr/bin/env bash # Restore an Eurskem AI backup created by backup.sh.
`deploy/ionos/setup_host.sh`	#!/usr/bin/env bash set -euo pipefail if [[${EUID} -ne 0]]; then echo "Run this once as root on a fresh Ubuntu 24.04 IONOS VPS." >&2 exit 1 fi deploy_user=${DEPLOY_USER:-eurskem-deploy} app_group=eurskem-app app_gid=10001 apt-get update apt-get install -y ca-certificates curl fail2ban rsync ufw install -m 0755 -d /etc/apt/keyrings curl -fsSL https://download.docker.com/linux/ubuntu/gpg \ -o /etc/apt/keyrings/docker.asc chmod a+r /etc/apt/keyrings/docker.asc .
`observability/grafana/dashboards/platform.json`	{ "title": "AWP Platform", "uid": "awp-platform", "schemaVersion": 39, "time": { "from": "now-1h", "to": "now" }, "panels": [{ "title": "Node latency p95 (s) by type", "type": "timeseries", "gridPos": { "h": 8, "w": 12, "x": 0, "y": 0 }, "targets": [{ "expr": "histogram_quantile(0.95, sum(rate(awp
`observability/grafana/provisioning/dashboards/awp.yml`	apiVersion: 1 providers: - name: awp folder: "" type: file options: path: /etc/grafana/provisioning/dashboards
`observability/prometheus.yml`	global: scrape_interval: 15s scrape_configs: - job_name: "awp-platform" metrics_path: /metrics static_configs: - targets: ["app:8000"]
`observability/prometheus/alerts.yml`	# Eurskem AI — minimum viable production alert rules.
`scripts/backfill_knowledge_resources.py`	Idempotent legacy -> Knowledge Studio resource backfill.
`scripts/build_documentation_pdf.py`	Generate the canonical 24-volume codebase documentation portfolio.
`scripts/build_eu_proposal_copy_bundle.py`	Build the copy-ready sourcebook and ZIP for this implementation.
`scripts/build_hitl_editor_bundle.py`	Build complete copy/paste files and a path-preserving HITL editor ZIP.
`scripts/build_workflow_file_inputs_bundle.py`	Build complete copy/paste files and a path-preserving ZIP.
`scripts/build_workflow_qa_matrix.py`	Build a complete, evidence-based QA matrix for every workflow YAML file.
`scripts/check_api_routes.py`	Fail fast when the Studio frontend and FastAPI route table disagree.
`scripts/debug_compressor.py`	Debug the compressor in isolation.
`scripts/generate_docstrings.py`	Generate missing docstrings across the Python codebase.
`scripts/generate_fixtures.py`	Generate three synthetic RFP fixtures for the Phase 2 test corpus.
`scripts/generate_production_env.py`	Generate every non-provider secret required by the IONOS Compose stack.
`scripts/generate_reference_workflows.py`	Generate 400 hidden, deterministic node-reference workflows.
`scripts/ingest_samples.py`	Repeatable Phase 2 ingestion: seed Weaviate from samples/.
`scripts/inspect_weaviate.py`	Inspect what's actually indexed in Weaviate.
`scripts/live_provider_smoke.py`	Explicit, low-cost provider smoke test for the protected manual CI job.
`scripts/load_test.py`	Authenticated 100-concurrent-user release gate with no LLM calls.
`scripts/manage_user.py`	Create or rotate a Mongo-backed local application user.
`scripts/new_node_type.py`	Scaffold one fully standardized workflow node module.
`scripts/openrouter_health_check.py`	Live health check for every LLM in OpenRouter's catalog.
`scripts/preflight_workflows.py`	Validate workflow YAML files without running nodes or calling an LLM.

File	Responsibility
`scripts/production_preflight.py`	Fail-fast validation for the IONOS production environment file.
`scripts/register_biomass.py`	Register biomass module.
`scripts/render_sample_proposal.py`	Render the checked-in Horizon proposal sample for visual QA.
`scripts/render_sample_proposal_docx.py`	Render the checked-in Horizon proposal DOCX sample for visual QA.
`scripts/retag_work_programme.py`	Retag work programme module.
`scripts/seed_collections.py`	Seed collections module.
`scripts/smoke_events.py`	Run a workflow with a tracing subscriber.
`scripts/smoke_llm.py`	Phase 3 smoke test — LLM gateway end-to-end with fallback routing.
`scripts/smoke_production.py`	Public TLS, readiness, and security-header smoke test.
`scripts/smoke_retrieval.py`	Phase 3 smoke test — full retrieval pipeline end-to-end.
`scripts/smoke_structured.py`	Phase 3 smoke test — structured output via the LLM gateway.
`scripts/start_snippet_runner_local.sh`	#!/usr/bin/env bash set -euo pipefail repo_root=$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd) socket_path=${SNIPPET_RUNNER_SOCKET_PATH:-/tmp/snippet-runner.sock} pid_file=${SNIPPET_RUNNER_PID_FILE:-/tmp/eurskem-snippet-runner.pid} log_file=${SNIPPET_RUNNER_LOG_FILE:-/tmp/eurskem-snippet-runner.log} python=${PYTHON:-"${repo_root}/.venv/bin/python"} if [[!
`scripts/verder_knowledge_config.json`	{ "owner_scope_id": "ayush", "collection_id": "col_01M0D8GYAD5FBWKTF7GCE1Z16F", "retrieval_profile_id": "retprof_01M0D8GYAQWKK3BSMK0A0384WG" }
`scripts/verder_process_emails.py`	Driver for the Verder Liquids assessment POC.
`scripts/verder_reingest_production.py`	Re-ingest the 7 Verder pump datasheets into col_01M0D8GYAD5FBWKTF7GCE1Z16F through the REAL production Knowledge Studio pipeline (IngestionJobRunner), so the collection's already-linked Parser/Chunking Profiles (vision_augmented, parent_child) are genuinely exercised and per-file `product` metadata actually lands in Weaviate — none of which happened under the original legacy ingest_file() seeding (scripts/verder_setup_knowledge.py).
`scripts/verder_setup_knowledge.py`	One-time setup for the Verder Liquids assessment POC's product knowledge.
`scripts/verify_knowledge_studio.py`	Live acceptance harness for Knowledge Studio.

Symbol	Kind	Location	Signature / summary
Settings	class	`app/config.py:15`	`BaseSettings` Provides the Settings behaviour.
Settings.workflow_file_max_bytes	method	`app/config.py:72`	`self` The workflow file max bytes.
Settings.workflow_file_max_total_bytes	method	`app/config.py:77`	`self` The workflow file max total bytes.
Settings.max_request_body_bytes	method	`app/config.py:82`	`self` The max request body bytes.
Settings.validate_production_security	method	`app/config.py:442`	`self` Refuse to boot production with development security settings.
Settings.moonshot_api_key	method	`app/config.py:560`	`self` Kimi web search and vision (app/tools/web_io.py,

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			vision_io.py) authenticate as the same Moonshot account as the LLM-gateway route — one credential, LOCAL_KIMI_API_KEY, not a second parallel setting.
Settings.kimi_api_base_url	method	`app/config.py:567`	`self` The kimi api base url.
Settings.required_readiness_services	method	`app/config.py:572`	`self` The required readiness services.
Settings.resolved_paper_search_mcp_path	method	`app/config.py:581`	`self` The resolved paper search mcp path.
Settings.resolved_scientific_skills_path	method	`app/config.py:587`	`self` The resolved scientific skills path.
Settings.allowed_scientific_skills	method	`app/config.py:592`	`self` The allowed scientific skills.
Settings.allowed_cors_origins	method	`app/config.py:597`	`self` The allowed cors origins.
Settings.allowed_hosts	method	`app/config.py:602`	`self` The allowed hosts.
_csv_values	function	`app/config.py:607`	`value: str` Internal helper for the csv values step.
_redis_url_has_password	function	`app/config.py:619`	`url: str` Return whether a Redis URL contains a non-empty password component.
_valid_http_url	function	`app/config.py:630`	`value: str` Internal helper for the valid http url step.
_is_placeholder	function	`app/config.py:653`	`value: str` Return whether placeholder.